

1. Calculate the molarity of a solution containing 5 g of NaOH in 450 mL solution (Ncert Text)

Ans: Given: weight=5g, V=450ml $\Rightarrow$ 0.450L, Molar mass of NaOH=23+16+1=40g mol⁻¹

$$n = \frac{5g}{40g\ mol^{-1}} \Rightarrow 0.125\text{mol} \quad M = n/(V(\text{in litre})) \Rightarrow M = 0.125\text{mol}/0.450\text{L}$$

$$\Rightarrow M = 0.278\text{mol L}^{-1} / 0.278\text{mol dm}^{-3}$$

2. Calculate molality of 2.5 g of ethanoic acid (CH₃COOH) in 75 g of benzene (Ncert Text)

Ans: Given: weight of solvent (Benzene)=75g=0.075Kg, weight solute ethanoic acid=2.5g,

Molar mass of CH₃COOH=12+3+12+32+1=60g mol⁻¹

$$n = \frac{2.5g}{60g\text{mol}^{-1}} \Rightarrow 0.0417\text{mol}, \quad m = \frac{n}{\text{solvent in Kg}} \Rightarrow m = \frac{0.0417\text{mol}}{0.075\text{Kg}} \Rightarrow m = 0.556\text{mol Kg}^{-1}$$

3. Calculate the mass percentage of benzene (C₆H₆) and carbon tetrachloride (CCl₄) if 22 g of benzene is dissolved in 122 g of carbon tetrachloride (NCERT)

Ans: Mass of Benzene = 22g mass of Carbon tetrachloride = 122 g

Mass of solution = mass of benzene + mass of carbon tetrachloride = 22+122=144g

$$\text{Mass \% of Benzene} = \frac{\text{Mass of benzene solution}}{\text{Total mass of the solution}} \times 100$$

$$\Rightarrow \frac{22g}{144g} \times 100 \Rightarrow 15.27\%$$

$$\text{Mass percentage of Carbo tetrachloride} = \frac{\text{Mass of carbon tetrachloride in the solution}}{\text{Total mass of the solution}} \times 100$$

$$\Rightarrow \frac{122g}{144g} \times 100 = 84.72\%$$

4. Calculate the mole fraction of urea (molar mass = 60g mol⁻¹) in aqueous solution containing 40% of urea by mass (NCERT)

Ans: mass of Urea= 40% $\Rightarrow$ 40g, mass of water = 60g, Molar mass of Urea=60g mol⁻¹,

Molar mass of Water=18g mol⁻¹ therefore no. of moles of urea

$$n_{\text{urea}} = \frac{\text{mass/weight of Urea}}{\text{molar /molecular of Urea}} \Rightarrow n_{\text{urea}} = \frac{40g}{60g/\text{mol}} \Rightarrow n_{\text{urea}} = 0.66\text{mol}$$

$$\text{no. of moles of water } n_{\text{water}} = \frac{\text{mass/weight of water}}{\text{molar /molecular of water}} \Rightarrow n_{\text{water}} = \frac{60g}{18g/\text{mol}} \Rightarrow n_{\text{water}} = 3.33\text{mol}$$

$$\text{mole fraction of urea} = \chi_{\text{urea}} = \frac{n_{\text{urea}}}{n_{\text{urea}} + n_{\text{water}}} \Rightarrow \chi_{\text{urea}} = \frac{0.66\text{mole}}{0.66\text{mole} + 3.33\text{mole}} \Rightarrow \chi_{\text{urea}} = 0.165$$

$$\text{mole fraction of water} = \chi_{\text{water}} = \frac{n_{\text{water}}}{n_{\text{urea}} + n_{\text{water}}} \Rightarrow \chi_{\text{water}} = \frac{3.33\text{mole}}{0.66\text{mole} + 3.33\text{mole}} \Rightarrow \chi_{\text{water}} = 0.834$$

$$\chi_{\text{urea}} + \chi_{\text{water}} = 0.165 + 0.834 = 0.999$$

5. Calculate the mole fraction of ethylene glycol (C₂H₆O₂) in a solution containing 20% of C₂H₆O₂ by mass (NCERT).

Ans: mass of ethylene glycol= 20g

mass of water=80 g

Molecular mass of ethylene glycol

C₂H₆O₂ = 12X2 + 1X6 + 16X2=62g mol⁻¹

Molecular mass of

H₂O = 1X2+16X1= 18g mol⁻¹

$$n_{\text{ethylene glycol}} = \frac{\text{mass of ethylene glycol}}{\text{molecular of ethylene glycol}}$$

$$\Rightarrow n_{\text{Ethylene glyco}} = \frac{20g}{62g/mol} \Rightarrow n_{\text{Ethylene glyco}} = 0.322 \text{ mol}$$

$$\text{no. of moles of water } n_{\text{water}} = \frac{\text{mass/weight of water}}{\text{molar /molecular of water}}$$

$$\Rightarrow n_{\text{water}} = \frac{80g}{18g/mol} \Rightarrow n_{\text{water}} = 4.44 \text{ mol}$$

$$\text{mole fraction of Ethylene glycol} = \chi_{\text{ethylene glycol}} = \frac{n_{\text{Ethylene glycol}}}{n_{\text{Ethylene glycol}} + n_{\text{water}}}$$

$$\Rightarrow \chi_{\text{ethylene glycol}} = \frac{0.322 \text{ mol}}{0.322 \text{ mol} + 4.44 \text{ mol}} \Rightarrow \chi_{\text{ethylene glycol}} = 0.0676$$

$$\text{mole fraction of water} = \chi_{\text{water}} = \frac{n_{\text{water}}}{n_{\text{Ethylene glycol}} + n_{\text{water}}} \Rightarrow \chi_{\text{water}} = \frac{4.44 \text{ mol}}{0.322 \text{ mol} + 4.44 \text{ mol}} \Rightarrow \chi_{\text{water}} = 0.932$$

$$\chi_{\text{ethylene glycol}} + \chi_{\text{water}} = 0.0676 + 0.932 = 0.999$$

6. Calculate the molarity of each of the following solutions: (a) 30 g of $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in 4.3 L of solution (b) 30 mL of 0.5 M H_2SO_4 diluted to 500 mL. (Ncert Text)

Ans: a) Given: weight=30g, V=4.3L,

$$\text{Molar mass of } \text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} = 291 \text{ g mol}^{-1}$$

$$M = \frac{w \times 1000}{m \times V} \Rightarrow M = \frac{30g \times 1000}{291g/mol \times 4.3 \times 1000} \Rightarrow M = 0.024 \text{ mol L}^{-1}$$

b) Given: $M_1 = 0.5 \text{ M}$ $V_1 = 30 \text{ ml}$ $V_2 = 500 \text{ ml}$, $M_2 = ?$

$$M_1 V_1 = M_2 V_2 \Rightarrow M_2 = \frac{M_1 V_1}{V_2} \Rightarrow$$

$$M_2 = \frac{0.5 \text{ M} \times 30 \text{ ml}}{500 \text{ ml}} \quad M_2 = 0.03 \text{ M}$$

7. Calculate (a) molality (b) molarity and (c) mole fraction of KI if the density of 20% (mass/mass) aqueous KI is 1.202 g mL^{-1} .

Ans: 20% (mass/mass) aqueous KI

= 20 g KI present in 100 g of solution = 20 g of KI present in 80 g of water (Solvent)

$$\text{Molar mass of KI} = 39 + 127 = 166 \text{ g mol}^{-1}$$

$$n_{\text{KI}} = \frac{\text{mass of KI}}{\text{molar mass of KI}} \Rightarrow n_{\text{KI}} = \frac{20 \text{ g}}{166 \text{ g/mol}} \Rightarrow n_{\text{KI}} = 0.1208 \text{ mol}$$

$$\text{a) Molality}(m) = \frac{n}{\text{mass of solvent in Kg}} = \frac{0.1208 \text{ mol}}{0.080 \text{ kg}} \quad \text{Molality}(m) = 1.51 \text{ mol/Kg} = 1.51 \text{ m}$$

b) density of solution(d) = 1.202 g mL^{-1} mass of solution = 100g

$$d = \frac{\text{mass}}{\text{volume}} \Rightarrow \text{volume} = \frac{\text{mass}}{\text{density}} \Rightarrow \text{volume} = \frac{100 \text{ g}}{1.202 \text{ g/mL}} \Rightarrow \text{volume} = 83.19 \text{ mL} \Rightarrow \text{volume} = 0.08319 \text{ L}$$

$$\text{Molarity}(M) = \frac{n}{\text{Volume of solution in Litre}} = \frac{0.1208 \text{ mol}}{0.08319 \text{ L}} \quad \text{Molarity}(M) = 1.45 \text{ M}$$

c) Moles of KI = 0.1208 mol

$$n_{\text{H}_2\text{O}} = \frac{\text{mass of water}}{\text{Molar mass of water}} \Rightarrow n_{\text{H}_2\text{O}} = \frac{80 \text{ g}}{18 \text{ g/mol}} \Rightarrow n_{\text{H}_2\text{O}} = 4.44 \text{ mol}$$

$$\Rightarrow \chi_{\text{KI}} = \frac{\text{moles of KI}}{\text{mole of KI} + \text{moles of water}}$$

$$\Rightarrow \chi_{\text{KI}} = \frac{0.1208 \text{ mol}}{0.1208 \text{ mol} + 4.44 \text{ mol}} \Rightarrow \chi_{\text{KI}} = 0.0263$$

Raoult's law

8. The vapour pressure of pure liquids 'A' and 'B' are 450 and 700 mm Hg respectively, at 350 K. Find out the composition of the liquid mixture if total vapour pressure is 600 mm Hg. (E1-25)

Ans: Given: $P_A^0 = 450 \text{ mmHg}$ $P_B^0 = 700 \text{ mmHg}$

$$P_{\text{Total}} = P_B^0 + (P_A^0 - P_B^0) \chi_A$$

$$600 \text{ mmHg} = 700 + (450 - 700) \text{ mmHg } \chi_A$$

$$250 \chi_A = 100 \Rightarrow \chi_A = \frac{100}{250} \Rightarrow \chi_A = 0.40$$

$$\chi_A + \chi_B = 1 \Rightarrow \chi_B = 1 - 0.40 \Rightarrow \chi_B = 0.60$$

9. Vapour pressure of compound 'A' (molar mass 120 g/mol) and compound 'B' (molar mass 85g/mol) at 298 K are 200 mm Hg and 415 mm Hg respectively. Calculate the vapour pressure of the solution prepared by mixing 26.0 g of compound 'A' and 40 g of compound 'B' at 298 K. (E2-24)

Ans:

$$\text{Mole fraction of compound B} (\chi_B) = \frac{n_B}{n_A + n_B} = \frac{\frac{40}{85}}{\frac{26}{120} + \frac{40}{85}} = \frac{0.47}{0.216 + 0.47} = 0.685$$

$$P_{\text{total}} = P_A^0 + (P_B^0 - P_A^0) \chi_B$$

$$= 200 + (415 - 200) 0.685 = 347.27 \text{ mmHg}$$

RELATIVE LOWERING OF VAPOUR PRESSURE

10. The vapour pressure of pure benzene at a certain temperature is 0.850 bar. A non-volatile, non-electrolyte solid weighing 0.5 g when added to 39.0 g of benzene (molar mass 78 g mol⁻¹). Vapour pressure of the solution, then, is 0.845 bar. What is the molar mass of the solid substance? (S-14, Ncert text, S-17)

Ans: Given: $p_1^0 = 0.850 \text{ mmHg}$; $p_1 = 0.845 \text{ mmHg}$,

$$M_1 = 78 \text{ g mol}^{-1}; \quad w_2 = 0.5 \text{ g}, \quad w_1 = 39 \text{ g}, \quad M_2 = ?$$

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 X M_1}{M_2 X w_1} \Rightarrow \frac{0.850 \text{ mmHg} - 0.845 \text{ mmHg}}{0.850 \text{ mmHg}} = \frac{0.5 \text{ g} \times 78 \text{ g mol}^{-1}}{M_2 \times 39.0 \text{ g}} \Rightarrow M_2 = 170 \text{ g/mol}$$

11. Vapour pressure of benzene is 200 mmHg. When 2 gram of a non volatile solute dissolved in 78 gram benzene. Benzene has vapour pressure of 195 mm Hg. Calculate the molar mass of the solute. [molar mass of benzene is 78 gram mol⁻¹ (A-20)]

Ans: $p_1^0 = 200 \text{ mmHg}$; $p_1 = 195 \text{ mmHg}$;

$$M_1 = 78 \text{ g mol}^{-1}; \quad w_2 = 2 \text{ g}, \quad w_1 = 78 \text{ g} \quad M_2 = ?$$

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 X M_1}{M_2 X w_1} \Rightarrow \frac{200 \text{ mmHg} - 195 \text{ mmHg}}{200 \text{ mmHg}} = \frac{2 \text{ g} \times 78 \text{ g mol}^{-1}}{M_2 \times 78 \text{ g}} \Rightarrow M_2 = 80 \text{ g/mol}$$

12. 5.8 gm of a non volatile solute was dissolved in 100g of carbon disulphide (molar mass=76). The vapour pressure of the solution was found to be 190mm Hg. Calculate the molecular mass of the solute. The vapour pressure of pure carbon disulphide(CS_2) 195mmHg (A-16, S-18)

Ans: $p_1^0 = 195\text{mmHg}$; $p_1 = 190\text{mmHg}$;

$$M_1 = 76 \text{ g mol}^{-1}; \quad w_2 = 5.8 \text{ g}; \quad w_1 = 100\text{g}, M_2 = ?$$

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 X M_1}{M_2 X w_1} \Rightarrow \frac{195\text{mmHg} - 190\text{mmHg}}{195\text{mmHg}} = \frac{5.8\text{g} \times 76\text{g mol}^{-1}}{M_2 \times 100\text{g}} \Rightarrow M_2 = 171.912 \text{ g/mol}$$

13. The vapour pressure of water is 12.3 kPa at 300 K. Calculate vapour pressure of 1 molal solution of a non-volatile solute in it (Ncert text)

Ans: Given: $p_1^0 = 12.3 \text{ kPa}$

1 molal solution = 1mole solute in 1000 g/1Kg of water

Molar mass of water = 18 g/mol

$$n_{\text{H}_2\text{O}} = \frac{\text{mass of water}}{\text{Molar mass of water}} \quad n_{\text{H}_2\text{O}} = \frac{1000 \text{ g}}{18 \text{ g/mol}} \Rightarrow 55.55\text{mol}$$

$$x_{\text{Solute}} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{H}_2\text{O}}} \Rightarrow x_{\text{Solute}} = \frac{1}{1 + 55.55}$$

$$\Rightarrow x_{\text{Solute}} = 0.0176$$

$$\frac{p_1^0 - p_1}{p_1^0} = x_{\text{Solute}} \Rightarrow \frac{12.3\text{kPa} - p_1}{12.3\text{kPa}} = 0.0176 \Rightarrow p_1 = 12.08$$

14. Vapour pressure of water at 293 K is 17.535 mm Hg. Calculate the vapour pressure of water at 293 K when 25 g of glucose is dissolved in 450 g of water (Ncert text)

Ans: Given: $p_1^0 = 17.535 \text{ mm Hg}$ $p_1 = ?$ $w_2 = 25\text{g}$ $w_1 = 450 \text{ g}$

molar mass of glucose($\text{C}_6\text{H}_{12}\text{O}_6$) $M_2 = 12 \times 6 + 1 \times 12 + 16 \times 6 = 180 \text{ g/mol}$

Molar mass of water (H_2O) $M_1 = 1 \times 2 + 16 \times 1 = 18 \text{ g/mol}$

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 X M_1}{M_2 X w_1} \Rightarrow \frac{17.535 - p_1}{17.535} = \frac{25 \text{ g} \times 18\text{g mol}^{-1}}{180 \text{ g mol}^{-1} \times 450\text{g}} \quad p_1 = 17.438 \text{ mm Hg}$$

Elevation in Boiling point

15. On dissolving 2.34 g of solute in 40g of benzene, the boiling point of solution was higher than that of benzene by 0.81K. K_b value of benzene is $2.53 \text{ K Kg mol}^{-1}$. Calculate the molar mass of the solute (A-14, S-20)

Ans: $W_2 = 2.34 \text{ gm}$, $w_1 = 40\text{gm}$, $K_b = 2.53 \text{ K kg mol}^{-1}$, $\Delta T_b = 0.81\text{K}$

$$M_2 = \frac{1000 \times w_2 \times K_b}{\Delta T_b \times w_1} \Rightarrow M_2 = \frac{1000 \times 2.34\text{g} \times \frac{2.53\text{K Kg}}{\text{mol}}}{0.81\text{K} \times 40\text{g}} \Rightarrow M_2 = 182.72\text{g/mol}$$

16. The boiling point of benzene is 353.23 K. When 1.80 g of a non-volatile solute was dissolved in 90 g of benzene, the boiling point is raised to 354.11 K. Calculate the molar mass of the solute. K_b for benzene is $2.53 \text{ K kg mol}^{-1}$ (Ncert text, S-15, A-18, E3-24)

Ans: $W_2 = 1.80\text{gm}$, $w_1 = 90\text{gm}$

$$T_b = 353.23\text{K} \quad T_b^0 = 354.11 \quad \Delta T_b = T_b - T_b^0 \Rightarrow T_b^0 = 354.11 \text{ K} - 353.23 \Rightarrow K = 0.88 \text{ K}$$

$$M_2 = \frac{1000 \times w_2 \times K_b}{\Delta T_b \times w_1} \Rightarrow M_2 = \frac{1000 \times 1.80\text{g} \times 2.53\text{K Kg/mol}}{0.88\text{K} \times 90\text{g}} \Rightarrow M_2 = 57.5\text{g/mol}$$

17. On dissolving 3.46g of solute in 100g of water, the boiling point of solution was raised to that of pure water by 0.12K. Calculate the molar mass of the non-volatile solute. (Given: K_b of water is 0.51Kkgmol^{-1}) (A-22)

Ans: $M_2 = \frac{1000 \times w_2 \times K_b}{\Delta T_b \times w_1} \Rightarrow M_2 = \frac{1000 \times 3.46 \times 0.51 \text{Kkg/mol}}{0.12 \text{K} \times 100 \text{g}} \Rightarrow M_2 = 147.05 \text{g mol}^{-1}$

18. 24 gm of a non volatile, non electrolyte solute is added to 600 g of water the boiling point of the resulting solution is 373.35K. calculate the molar mass of the solute (given boiling point of pure water = 373 K and k_b for water = 0.52Kkgmol^{-1})

Ans: $W_2 = 24 \text{ gm}$, $w_1 = 600 \text{ gm}$, $T_b = 373 \text{ K}$, $T_b^0 = 373.35$, $\Delta T_b = 373.35 - 373 = 0.35 \text{ K}$, $K_b = 0.52 \text{Kkgmol}^{-1}$

$$M_2 = \frac{1000 \times w_2 \times K_b}{\Delta T_b \times w_1} \Rightarrow M_2 = \frac{1000 \times 24 \times 0.52 \text{Kkg/mol}}{0.35 \text{K} \times 600 \text{g}} \Rightarrow M_2 = 59.42 \text{g mol}^{-1}$$

19. Boiling point of water at 750 mm Hg is 99.63°C . How much sucrose is to be added to 500 g of water such that it boils at 100°C (Ncert text)

Ans: Given: $W_2 = ?$, $w_1 = 500 \text{ gm}$

$$T_b = 100^\circ\text{C}, \quad T_b^0 = 99.63^\circ\text{C} \Rightarrow \Delta T_b = T_b - T_b^0 \Rightarrow \Delta T_b = 100^\circ\text{C} - 99.63^\circ\text{C} \Rightarrow \Delta T_b = 0.37 \text{ K}$$

Molar mass of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) $M_2 = (12 \times 12 + 1 \times 22 + 16 \times 11) = 342 \text{ g/mol}$

$$M_2 = \frac{1000 \times w_2 \times K_b}{\Delta T_b \times w_1} \Rightarrow W_2 = \frac{\Delta T_b \times w_1 \times M_2}{K_b \times 1000}$$

$$W_2 = \frac{0.37 \text{ K} \times 500 \text{ g} \times 342 \text{ g/mol}}{0.52 \text{ K kg/mol} \times 1000} \Rightarrow W_2 = 121.6 \text{ g}$$

20. 18 g of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is dissolved in 1 kg of water in a saucepan. At what temperature will water boil at 1.013 bar? K_b for water is $0.52 \text{ K kg mol}^{-1}$

Ans: Moles of glucose = $\frac{\text{mass of glucose}}{\text{molar mass of glucose}} = \frac{18 \text{ g}}{180 \text{ g mol}^{-1}} = 0.1 \text{ mol}$

Number of kilograms of solvent = 1 kg

$$\text{molality of glucose solution} = \frac{\text{moles of glucose}}{\text{mass of solvent}} = \frac{0.1 \text{ mol}}{1 \text{ Kg}} = 0.1 \text{ mol kg}^{-1}$$

For water, change in boiling point

$$\Delta T_b = K_b \times m = 0.52 \text{ K kg mol}^{-1} \times 0.1 \text{ mol kg}^{-1} = 0.052 \text{ K}$$

Since water boils at **373.15 K** at 1.013 bar pressure, therefore, the boiling point of solution will be **$373.15 + 0.052 = 373.202 \text{ K}$** .

Depression in freezing point

21. A solution containing 18g of non volatile, non electrolyte solute dissolved in 200g of water freezes at 272.07K. calculate the molecular mass of solute. Given: $K_f = 1.86 \text{ K kg/mol}$, freezing point of water = 273K) (A-15)

Ans: $W_2 = 18 \text{ g}$, $w_1 = 200 \text{ g}$, $K_f = 1.86 \text{Kkgmol}^{-1}$, $T_f = 272.0 \text{ K}$, $T_f^0 = 273 \text{ K} \Rightarrow \Delta T_f = T_f^0 - T_f \Rightarrow \Delta T_f = 273 \text{ K} - 272.07 \text{ K} \Rightarrow \Delta T_f = 0.93 \text{ K}$

$$\Delta T_f = \frac{1000 \times w_2 \times K_f}{M_2 \times w_1} \Rightarrow M_2 = \frac{1000 \times w_2 \times K_f}{\Delta T_f \times w_1} \Rightarrow M_2 = \frac{1000 \times 18 \times 1.86 \text{Kkg/mol}}{0.93 \text{K} \times 200 \text{g}} \Rightarrow M_2 = 180 \text{g/mol}$$

22. 31 g of a unknown molecular material is dissolved in 500 g of water. The resulting solution freezes at 271.14K. Calculate the molar mass of the material[Given : K_f for water= 1.86K kg mol^{-1} , T_f^0 water= 273K] (A-19,S-22)

Ans: $W_2=31\text{g}$, $w_1=500\text{g}$, $K_f=1.86\text{K kg mol}^{-1}$, $T_f=271.14\text{K}$, $T_f^0=273\text{K} \Rightarrow \Delta T_f=T_f^0 - T_f \Rightarrow \Delta T_f=273\text{K}-271.14\text{K} \Rightarrow \Delta T_f=1.86\text{K}$

$$\Delta T_f = \frac{1000 \times w_2 \times K_f}{M_2 \times w_1} \Rightarrow M_2 = \frac{1000 \times w_2 \times K_f}{\Delta T_f \times w_1} \Rightarrow M_2 = \frac{1000 \times 31\text{g} \times 1.86\text{K kg/mol}}{1.86\text{K} \times 500\text{g}} \Rightarrow M_2 = 62\text{g/mol}$$

23. 1.0g of a non-electrolyte solute dissolved in 50 g of benzene lowered the freezing point of benzene by 0.40 K. The freezing point depression constant of benzene is $5.12\text{ K kg mol}^{-1}$. Find the molar mass of the solute (NCERT TEXT, A-17, S-20,23)

Ans: $W_2=1.00\text{g}$, $w_1=50\text{g}$, $\Delta T_f=0.40\text{K}$ $K_f=5.12\text{K kg mol}^{-1}$

$$M_2 = \frac{1000 \times w_2 \times K_f}{\Delta T_f \times w_1} \Rightarrow M_2 = \frac{1000 \times 1.0\text{g} \times 5.12\text{K kg/mol}}{0.40\text{K} \times 50\text{g}} \Rightarrow M_2 = 256\text{g/mol}$$

24. 45 g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) is mixed with 600 g of water. Calculate (a) the freezing point depression and (b) the freezing point of the solution (NCERT)

Ans: Moles of ethylene glycol = $\frac{\text{mass of ethylene glycol}}{\text{molar mass of ethylene glycol}} = \frac{45\text{g}}{62\text{g mol}^{-1}} = 0.73\text{ mol}$

Mass of water in Kg = $600\text{g} = 0.6\text{ Kg}$

$$\text{molality of Ethylene glycol} = \frac{\text{moles of ethylene glycol}}{\text{mass of water in Kg}} = \frac{0.73\text{ mol}}{0.6\text{ Kg}} = 1.2\text{ mol kg}^{-1}$$

freezing point depression,

$$\Delta T_f = K_f m \Rightarrow \Delta T_f = 1.86\text{ K kg mol}^{-1} \times 1.2\text{ mol kg}^{-1} \Rightarrow \Delta T_f = 2.2\text{ K}$$

$$\text{Freezing point of the aqueous solution} = \Delta T_f = T_f^0 - T_f \Rightarrow 2.2\text{K} = 273.15 - T_f \\ \Rightarrow 2.2\text{K} - 273.15 = -T_f \Rightarrow 270.95\text{ K}$$

25. 300cm^3 of an aqueous solution of protein contain 2.12 g of protein, the osmotic pressure of such solution at 300K is found to be $3.89 \times 10^{-3}\text{ bar}$. Calculate the molar mass of the solution of the protein ($R=0.0823\text{ L bar mol}^{-1}\text{ K}^{-1}$) (S-16)

Ans: $w_2=2.12\text{g}$ $\pi = 3.89 \times 10^{-3}\text{ bar}$, $V = 300\text{ cm}^3 = 0.300\text{ litre}$ $T = 300\text{ K}$

$R = 0.083\text{ L bar mol}^{-1}\text{ K}^{-1}$

$$M_2 = \frac{w_2 R T}{\pi V} \Rightarrow M_2 = \frac{2.12\text{g} \times 0.0823\text{L bar mol}^{-1}\text{K}^{-1} \times 300\text{K}}{3.89 \times 10^{-3}\text{ bar} \times 0.300\text{ L}} \Rightarrow M_2 = 44852.4\text{ g/mol}$$

26. 200 cm^3 of an aqueous solution of a protein contains 1.26 g of the protein. The osmotic pressure of such a solution at 300 K is found to be $2.57 \times 10^{-3}\text{ bar}$. Calculate the molar mass of the protein.(Ncert Text)

Ans: $w_2=1.26\text{ g}$ $V = 200\text{cm}^3 = 0.200\text{ litre}$ $M_2=?$ $T = 300\text{ K}$ $R=0.0821\text{L atm mol}^{-1}\text{K}^{-1}$

$$M_2 = \frac{w_2 R T}{\pi V} \Rightarrow M_2 = \frac{1.26\text{g} \times 0.083\text{L bar mol}^{-1}\text{K}^{-1} \times 300\text{K}}{2.57 \times 10^{-3}\text{ bar} \times 0.200\text{ L}} \Rightarrow M_2 = 61.022\text{ g/mol}$$

27. 450cm^3 of an aqueous solution of a protein contains 1.0g of the protein. The osmotic pressure of such a solution at 310K is found to be $3.1 \times 10^{-4}\text{ bar}$. Calculate the molar mass of the protein. ($R = 0.083\text{L bar mol}^{-1}\text{ K}^{-1}$)(A-23)

$$M_2 = \frac{w_2 R T}{\pi V} \Rightarrow M_2 = \frac{1.0g \times 0.083L \text{ bar mol}^{-1}K^{-1} \times 310K}{3.1 \times 10^{-4} \text{ bar} \times 0.450L} \Rightarrow M_2 = 1,84,444 \text{ g/mol}$$

28. Calculate the osmotic pressure in Pascals exerted by a solution prepared by dissolving 1.0 g of polymer of molar mass 185000 in 450 mL of water at 37 °C [R= 8.314 × 10³ Pa L K⁻¹ mol⁻¹] (E1-25)

Ans: $w_2 = 1.0 \text{ g}$ $V = 450 \text{ mL} = 0.450 \text{ L}$ $M_2 = 185000$ $T = 37^\circ\text{C} = 37^\circ + 273 = 310 \text{ K}$

$$R = 8.314 \times 10^3 \text{ Pa L mol}^{-1}K^{-1}$$

$$\pi = \frac{w_2 R T}{M_2 V} \Rightarrow \pi = \frac{1.0 \text{ g} \times 8.314 \times 10^3 \text{ Pa L K}^{-1} \text{ mol}^{-1} \times 310 \text{ K}}{185000 \text{ mol}^{-1} \times 0.450 \text{ L}} \Rightarrow \pi = 0.03095 \times 10^3 \text{ Pa} \text{ OR } 30.95 \text{ Pa}$$

Henry's law

29. Henry's law constant for the molality of methane in benzene at 298 K is 4.27 × 10⁵ mm Hg. Calculate the solubility of methane in benzene at 298 K under 760 mm Hg. (P-24, Ncert text)

Ans: Given: $K_H = 4.27 \times 10^5 \text{ mm Hg}$, $P = 760 \text{ mm Hg}$

$$p = K_H x_{\text{Methane}} \Rightarrow x_{\text{Methane}} = \frac{p}{K_H}$$

$$x_{\text{Methane}} = \frac{760 \text{ mm Hg}}{4.27 \times 10^5 \text{ mm Hg}} \quad x_{\text{Methane}} = 1.78 \times 10^{-3}$$

30. A solution containing 30 g of a nonvolatile solute in 90 g water has vapour pressure 2.8 kPa at 298 K. At the same temperature vapour pressure of pure water is 3.64 kPa. Calculate the molar mass of solute

Ans: Given: $p_1^0 = 3.64 \text{ kPa}$; $p_1 = 2.8 \text{ kPa}$;

$$M_1 = 18 \text{ g mol}^{-1}; w_2 = 30 \text{ g}, w_1 = 90 \text{ g} \quad M_2 = ?$$

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 x M_1}{M_2 x w_1} \Rightarrow$$

$$\frac{3.64 \text{ kPa} - 2.8 \text{ kPa}}{3.64 \text{ kPa}} = \frac{30 \text{ g} \times 18 \text{ g mol}^{-1}}{M_2 \times 90 \text{ g}} \Rightarrow M_2 = 11.56 \text{ g/mol}$$

31. The vapour pressure of water is 12.3 kPa at 300 K. Calculate vapour pressure of 1 molal solution of a non-volatile solute in it (Ncert text)

Ans: Given: $p_1^0 = 12.3 \text{ kPa}$

$$1 \text{ molal solution} = 1 \text{ mole solute in } 1000 \text{ g/1 Kg of water}$$

$$\text{Molar mass of water} = 18 \text{ g/mol}$$

$$n_{H_2O} = \frac{\text{mass of water}}{\text{Molar mass of water}} \quad n_{H_2O} = \frac{1000 \text{ g}}{18 \text{ g/mol}} \Rightarrow 55.55 \text{ mol}$$

$$x_{\text{Solute}} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{H_2O}} \Rightarrow x_{\text{Solute}} = \frac{1}{1 + 55.55}$$

$$\Rightarrow x_{\text{Solute}} = 0.0176$$

$$\frac{p_1^0 - p_1}{p_1^0} = x_{\text{Solute}} \Rightarrow \frac{12.3 \text{ kPa} - p_1}{12.3 \text{ kPa}} = 0.0176 \Rightarrow p_1 = 12.08$$

32. Calculate the mass of a non-volatile solute (molar mass 40 g mol⁻¹) which should be dissolved in 114 g octane to reduce its vapour pressure to 80%.

Ans: **Given:** reduce its vapour pressure to 80%.=

$$p_1 = 0.8p_1^0$$

$$M_2 = 40 \text{ g/mol} \quad w_2 = ? \quad w_1 = 114 \text{ g}$$

$$\text{Molar mass of Octane (C}_8\text{H}_{18}\text{)} = 12 \times 8 + 1 \times 18 = 114 \text{ g/mol}$$

$$M_1 = 114 \text{ g/mol}$$

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 \times M_1}{M_2 \times w_1} \Rightarrow \frac{p_1^0 - 0.8p_1^0}{p_1^0} = \frac{w_2 \times 114 \text{ g mol}^{-1}}{40 \text{ g mol}^{-1} \times 114 \text{ g}}$$

$$1 - 0.8 = \frac{w_2}{40 \text{ g}} \Rightarrow w_2 = 8 \text{ g}$$

33. Vapour pressure of water at 293 K is 17.535 mm Hg. Calculate the vapour pressure of water at 293 K when 25 g of glucose is dissolved in 450 g of water (**Ncert text**)

Ans: **Given:** $p_1^0 = 17.535 \text{ mm Hg}$ $p_1 = ?$ $w_2 = 25 \text{ g}$ $w_1 = 450 \text{ g}$

$$\text{molar mass of glucose (C}_6\text{H}_{12}\text{O}_6\text{)}$$

$$M_2 = 12 \times 6 + 1 \times 12 + 16 \times 6 = 180 \text{ g/mol}$$

$$\text{Molar mass of water (H}_2\text{O)} \quad M_1 = 1 \times 2 + 16 \times 1 = 18 \text{ g/mol}$$

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 \times M_1}{M_2 \times w_1} \Rightarrow \frac{17.535 - p_1}{17.535} = \frac{25 \text{ g} \times 18 \text{ g mol}^{-1}}{180 \text{ g mol}^{-1} \times 450 \text{ g}}$$

$$p_1 = 17.438 \text{ mm Hg}$$